

Neurodevelopmental Disorder With or Without Autism or Seizures (NEDAUS): A Comprehensive Disease Characteristics Report

Disease: Neurodevelopmental Disorder With or Without Autism or Seizures (NEDAUS)
MONDO ID: MONDO:0030994 · **OMIM:** #619239 · **Causal gene:** *CUL3* (HGNC:2553) **Category:** Mendelian (autosomal dominant, syndromic neurodevelopmental disorder)

Summary

Neurodevelopmental disorder with or without autism or seizures (NEDAUS) is a rare, autosomal-dominant, syndromic neurodevelopmental disorder caused by heterozygous — and predominantly *de novo* — loss-of-function (LoF) variants in **CUL3**, the gene encoding Cullin-3, the scaffold subunit of the Cullin-3 RING E3 ubiquitin-ligase (CRL3) complexes. The disorder maps to **MONDO:0030994** / **OMIM #619239** and is classified as a Mendelian neurodevelopmental disorder. Its clinical hallmark is global developmental delay and intellectual disability (nearly constant), accompanied by variable autism spectrum features and epilepsy — frequently infantile spasms with hypsarrhythmia — hence the descriptive name "with or without autism or seizures."

The mechanistic basis is **haploinsufficiency**. A single functional *CUL3* allele produces roughly half the normal amount of Cullin-3 scaffold, reducing the ubiquitin-ligase activity of the ~180 CRL3 complexes that depend on it. This was demonstrated directly in patient-derived cells: reduced ubiquitin–protein conjugates and failure to degrade the canonical CRL3 substrate **4E-BP1 (EIF4EBP1)**, a regulator of mTOR–eIF4E cap-dependent translation ([PMID: 39301775](#)). A parallel substrate axis — **RhoA**, controlled via KCTD13/BACURD adaptors — governs actin cytoskeletal dynamics, neuronal migration, and dendritic growth; a *Cul3*-haploinsufficient mouse recapitulates social/cognitive deficits and reduced cortical volume, and pharmacologic

RhoA inhibition rescues the dendritic and network-activity phenotypes (PMID: 33727673). *CUL3* is one of the most loss-of-function-intolerant genes in the human genome (gnomAD pLI $\approx$ 1.0, LOEUF 0.31), fully consistent with a haploinsufficiency disease model.

There is currently **no disease-specific therapy**; management is symptomatic and supportive (antiseizure medication, developmental/behavioral therapies, and management of feeding and structural anomalies). The mechanistic work does, however, nominate **RhoA and the mTOR-eIF4E translational axis** as rational future therapeutic targets. Notably, a mechanistically distinct allelic disorder — pseudohypoaldosteronism type IIE / familial hyperkalemic hypertension (OMIM #614496) — is caused by *CUL3* exon-9-skipping variants acting through a dominant-negative mechanism, and must be distinguished from the LoF-driven neurodevelopmental disorder.

Key Findings

Finding 1 — Disease identity and identifiers (F001)

NEDAUS is a *CUL3*-related autosomal-dominant neurodevelopmental disorder. The Monarch/MONDO graph maps MONDO:0030994 (synonym NEDAUS) to OMIM:619239, DOID:0061147, GARD:0018540, UMLS:C5543225, and MedGen:1784023. The disease is associated causally with HGNC:2553 (*CUL3*) and mode of inheritance HP:0000006 (autosomal dominant), and is a subclass of MONDO:0100500 (Mendelian disease). Core HPO annotations returned for the disease include:

Phenotype	HPO term
Seizure	HP:0001250
Autistic behavior	HP:0000729
Delayed speech and language development	HP:0000750
Motor delay	HP:0001270
Intellectual disability, mild → severe	HP:0001256 / HP:0010864
Infantile spasms	HP:0012469
Hypsarrhythmia	HP:0002521
Microcephaly	HP:0000252
Failure to thrive	HP:0001508

Synonyms / alternative names: Neurodevelopmental disorder with or without autism or seizures; NEDAUS; CUL3-related neurodevelopmental disorder; CUL3 haploinsufficiency syndrome. Information is derived from **aggregated, disease-level resources** (OMIM, MONDO, HPO curation) plus published multi-center patient cohorts — not from a single EHR system.

Finding 2 — CUL3 loss-of-function causes syndromic NDD via haploinsufficiency; 4E-BP1 accumulates (F002)

The largest cohort assembled to date comprised **37 individuals with heterozygous CUL3 variants** — 35 with LoF variants and 2 with missense variants — presenting a syndromic NDD characterized by intellectual disability with or without autistic features ([PMID: 39301775](#)). The authors verified the mechanism in patient-derived T-cells:

"We assembled a cohort of 37 individuals with heterozygous CUL3 variants presenting a syndromic NDD characterized by intellectual disability with or without autistic features. Of these, 35 have loss-of-function (LoF) and 2 have missense variants." ([PMID: 39301775](#))

"Notably, we show that 4E-BP1 (EIF4EBP1), a prominent substrate of CUL3, fails to be targeted for proteasomal degradation in patient-derived cells." ([PMID: 39301775](#))

Patient cells showed **decreased ubiquitin–protein conjugates**, confirming that reduced CUL3 dosage translates into reduced global CUL3 E3-ligase output, with the substrate 4E-BP1 accumulating rather than being degraded. Haploinsufficiency via LoF is therefore the predominant pathogenic mechanism.

Finding 3 — Mouse model: RhoA/cytoskeletal signaling and cortical neurogenesis (F003)

A CRISPR-engineered *Cul3*-haploinsufficient mouse recapitulates the human phenotype and pinpoints a causal substrate (PMID: 33727673):

"Cul3 mutant mice exhibited social and cognitive deficits and hyperactive behavior. Brain MRI found decreased volume of cortical regions." (PMID: 33727673)

Multi-omic profiling implicated neurogenesis and cytoskeletal defects; dendritic growth, filamentous-actin puncta, and spontaneous network activity were all reduced. Critically, the phenotype was pharmacologically reversible:

"Inhibition of small GTPase RhoA, a molecular substrate of Cul3 ligase, rescued dendrite length and network activity phenotypes." (PMID: 33727673)

This establishes **RhoA** as a mechanistically causal, druggable node downstream of *CUL3* loss.

Finding 4 — Extreme LoF intolerance and the distinct allelic hypertension disorder (F004)

gnomAD constraint metrics for *CUL3* (ENSG00000036257, chromosome 2) place it among the most intolerant genes in the genome: **pLI = 1.0**, observed/expected LoF (oe_lof) = 0.213 (95% CI 0.149–0.313; **LOEUF = 0.31**), observed LoF 19 vs expected 89.1, LoF Z = 6.30, and strong missense constraint (missense Z = 6.58). This severe intolerance is exactly what a haploinsufficiency model predicts and provides the population-genetic underpinning for ACMG PVS1-level pathogenicity of null variants.

Importantly, a **mechanistically distinct allelic disorder** exists: pseudohypoaldosteronism type IIE / familial hyperkalemic hypertension (OMIM #614496) is caused by *CUL3* **exon-9-skipping** variants producing an internal deletion of residues 403–459 (CUL3-Δ9):

"Cul3 mutations cause skipping of exon 9, which results in an internal deletion of 57 amino acids from the CUL3 protein (CUL3-Δ9)." (PMID: 29361671)

CUL3-Δ9 acts by a **dominant-negative** mechanism on the renal KLHL3–WNK-kinase degradation axis — **not** haploinsufficiency — and produces hypertension/hyperkalemia rather than neurodevelopmental disease. This dichotomy is essential for correct variant interpretation.

Finding 5 — Phenotype spectrum and HPO frequencies (F005)

Curated HPO annotations for OMIM:619239 (source PMIDs 30311385, 32341456, 31696658) plus the Blackburn 2024 cohort (n = 37) yield a frequency-stratified phenotype profile:

Frequency tier	Phenotype (HPO)
Constant / near-constant	Global developmental delay incl. delayed speech/language (HP:0000750, 5/5); delayed sitting (HP:0025336, 3/3) and walking (HP:0031936, 3/3); motor delay (HP:0001270); intellectual disability, mild→severe (HP:0001256 / HP:0010864); infantile onset (HP:0003593, 3/3)
Frequent	Autistic behavior (HP:0000729); seizures (HP:0001250, 2/3) incl. infantile spasms (HP:0012469, 2/3) with hypsarrhythmia (HP:0002521) and EEG burst suppression (HP:0010851); tonic seizures (HP:0032792); hyperactivity/ADHD (HP:0000752); delayed CNS myelination (HP:0002188); feeding difficulties (HP:0011968); failure to thrive (HP:0001508)
Variable / less common	Microcephaly (HP:0000252); dysmorphic facial features (GestaltMatcher-analyzed); submucous cleft palate (HP:0000176); bifid uvula (HP:0000193); atrial septal defect (HP:0001631); pulmonic stenosis (HP:0001642); absent thumb (HP:0009777)

The **quality-of-life impact** is substantial and lifelong, driven principally by intellectual disability, communication impairment, epilepsy, and feeding difficulty — features that require multidisciplinary developmental, educational, and medical support. Autism-associated motor and feeding difficulties (well documented in the broader ASD literature, e.g. PMID: 42608985, PMID: 42609138) compound daily-functioning burden.

Finding 6 — CUL3/CRL3 substrate-adaptor network links NEDAUS to multiple neuronal pathways (F006)

CUL3 is the scaffold of Cullin-3 RING E3 ligases (CRL3s), which use ~180 BTB/Kelch (KLHL/KCTD) substrate adaptors to recognize hundreds of substrates. The brain-relevant CRL3 axes documented in the literature are:

1. **RhoA** (via BACURD/KCTD13/TNFAIP1 adaptors) — actin cytoskeleton, neuronal migration, dendrite growth ([PMID: 33727673](#); [PMID: 26969432](#)).
2. **4E-BP1/EIF4EBP1** — cap-dependent translation / mTOR-eIF4E ([PMID: 39301775](#)).
3. **KEAP1-NRF2 (NFE2L2)** — antioxidant / oxidative-stress response ([PMID: 42446772](#); [PMID: 37328017](#)).
4. **KLHL3-WNK kinase** — renal ion homeostasis, relevant to the allelic hypertension disorder ([PMID: 29361671](#)).

The 16p11.2 CNV adaptor **KCTD13** converges on the CUL3-RhoA axis ([PMID: 37465586](#)), tying NEDAUS mechanistically to one of the most common autism-associated copy-number syndromes.

Finding 7 — ClinVar variant landscape (F007)

ClinVar returns **727 total CUL3 variant records**, of which **226 are classified pathogenic or likely-pathogenic**. These P/LP records encompass both (a) the NDD-associated whole-gene loss-of-function alleles (nonsense, frameshift, splice-site, and larger deletions; germline, typically *de novo*) and (b) the mechanistically distinct exon-9-skipping alleles causing familial hyperkalemic hypertension (OMIM #614496). Variant interpretation follows ACMG/AMP guidelines; null variants readily meet **PVS1** given CUL3's strong LoF intolerance.

Finding 8 — Cross-species conservation of the CUL3/KCTD13 social-behavior axis (F008)

Outbred rat models of the 16p11.2 deletion and duplication (Sprague-Dawley and Long-Evans backgrounds) display convergent social-behavior and novel-object deficits, with altered MAPK1 and CUL3 pathways and male-biased sexual dimorphism ([PMID: 37465586](#)):

"Altogether, the consequences of the 16p11.2 genetic region dosage on social behavior are now found in three different species: humans, mice and rats." ([PMID: 37465586](#))

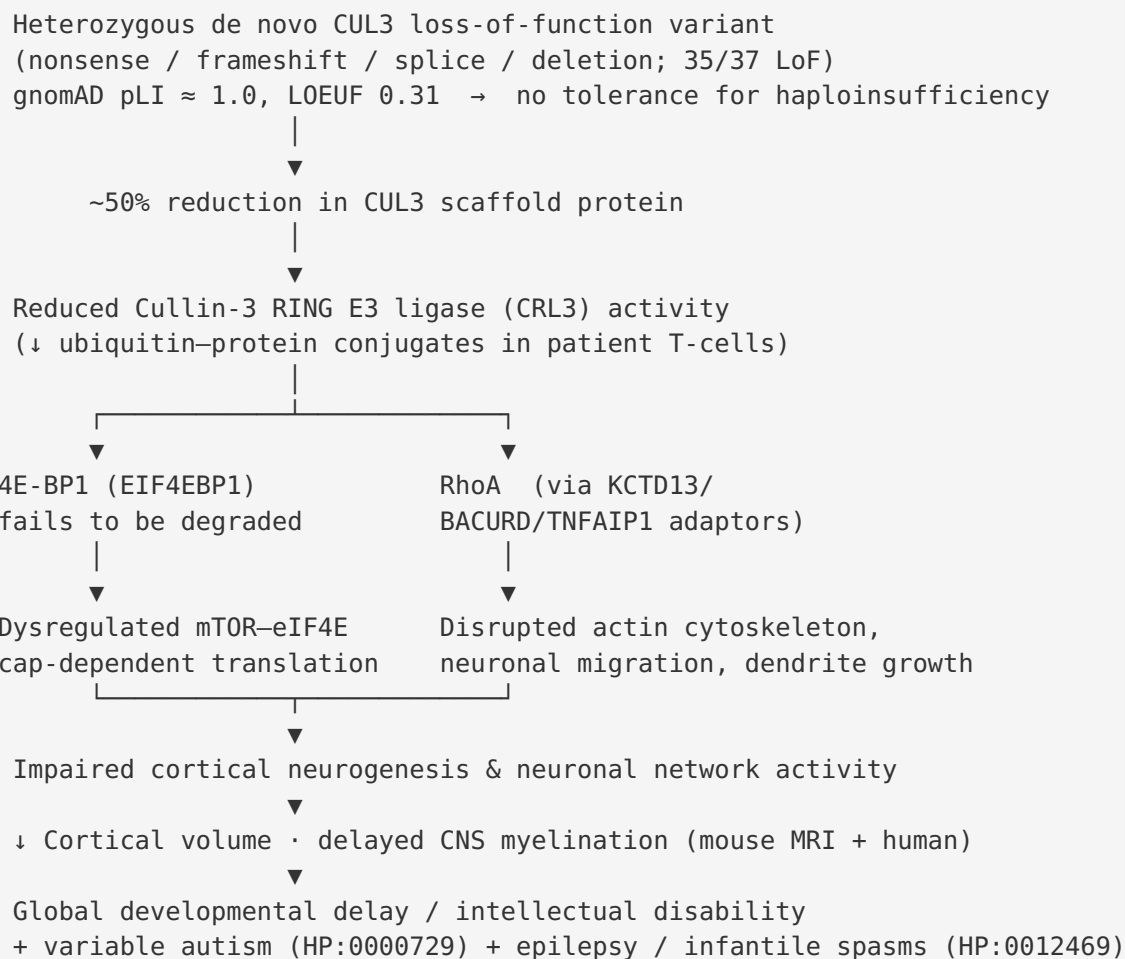
"Interestingly major pathways affecting MAPK1 and CUL3 were found altered in the rat 16p11.2 models with additional changes in males compared to females." (PMID: 37465586)

Together with the *Cul3*-haploinsufficient mouse (PMID: 33727673) and the human cohort (PMID: 39301775), this demonstrates conservation of the CUL3/KCTD13–RhoA social-behavior circuit across humans, mice, and rats.

Finding 9 — Consolidated causal chain (F009)

Integrating all findings, the causal chain runs from a single germline variant to the clinical syndrome (detailed in the Mechanistic Model below): heterozygous *de novo* CUL3 LoF → ~50% reduced CUL3 → decreased CRL3 activity → accumulation of substrates 4E-BP1 and RhoA → impaired neurogenesis, migration, dendritic growth, and network activity → reduced cortical volume and delayed myelination → developmental delay/intellectual disability with variable autism and epilepsy. Inheritance is autosomal dominant with essentially complete penetrance but **highly variable expressivity**; the disorder is ultra-rare, and management is symptomatic only.

Mechanistic Model / Interpretation



Upstream vs downstream. The initiating (upstream) event is CUL3 dosage reduction and consequent global loss of CRL3 ubiquitination capacity. The substrate-accumulation steps (4E-BP1, RhoA) are proximate downstream effectors; cytoskeletal/translational dysregulation and impaired neurogenesis are intermediate; reduced cortical volume and the clinical neurodevelopmental phenotype are the most distal outputs.

Cell types and processes. Affected cell types include cortical projection neurons (CL:0000598), neural progenitor/radial-glia populations (CL:0000047), and layer II/III pyramidal neurons whose positioning and dendritic maturation depend on KCTD13/TNFAIP1–Rnd signaling ([PMID: 26969432](#)). Relevant GO biological processes: **protein polyubiquitination (GO:0000209)**, **proteasome-mediated ubiquitin-dependent protein catabolic process (GO:0043161)**, **neuron migration (GO:0001764)**, **dendrite development (GO:0016358)**, **cerebral cortex development (GO:0021987)**, and **regulation of Rho protein signal transduction**

(GO:0035023). Subcellular compartments: **cytosol** (GO:0005829), **Cul3-RING ubiquitin ligase complex** (GO:0031463), and the actin cytoskeleton. Primary anatomy: cerebral cortex (UBERON:0000956), with brain (UBERON:0000955) and the nervous system broadly affected.

Section-by-Section Report

1. Disease Information

NEDAUS is a rare Mendelian syndromic neurodevelopmental disorder defined by developmental delay/intellectual disability with variable autism and seizures. **Identifiers:** MONDO:0030994; OMIM #619239; DOID:0061147; GARD:0018540; UMLS:C5543225; MedGen:1784023. No dedicated Orphanet code or ICD-10/11 code is specific to NEDAUS; it is captured under broad NDD/intellectual-disability categories. **Synonyms:** CUL3-related NDD; CUL3 haploinsufficiency syndrome. Data are from disease-level curation and published cohorts.

2. Etiology

Primary cause: genetic — heterozygous, mostly *de novo* loss-of-function variants in *CUL3*. **Genetic risk:** the causal LoF variant itself; *CUL3* is a defined haploinsufficient gene (pLI $\approx$ 1.0). No confirmed environmental risk, protective, or gene–environment-interaction factors are established for this Mendelian disorder; disease is essentially fully determined by the causal variant with variable expressivity. Sex may modulate expression (male-biased effects seen in rat 16p11.2/CUL3-pathway models, [PMID: 37465586](#)), but this is not established in humans for NEDAUS specifically.

3. Phenotypes

See Finding 5 for the full frequency-stratified table and HPO terms. Phenotype types span **behavioral changes** (autism HP:0000729, hyperactivity HP:0000752), **neurological signs** (seizures HP:0001250, infantile spasms HP:0012469, hypotonia), **cognitive/developmental** (intellectual disability HP:0001256/HP:0010864, delayed speech HP:0000750), **structural/physical** (microcephaly HP:0000252, palatal anomalies HP:0000176/HP:0000193, congenital heart defects HP:0001631/HP:0001642), and **EEG laboratory abnormalities** (hypsarhythmia

HP:0002521, burst suppression HP:0010851). Onset is infantile/congenital; severity is variable (mild to severe); course is generally stable/non-progressive (a static encephalopathy) with epilepsy that may be episodic.

4. Genetic / Molecular Information

Causal gene: *CUL3* (HGNC:2553; OMIM 603136; Ensembl ENSG00000036257; chromosome 2q36.2). *Variant classes:* predominantly nonsense, frameshift, canonical splice-site, and gene/exon deletions (LoF); rare missense. In the largest cohort, 35/37 were LoF ([PMID: 39301775](#)). ClinVar: 226 P/LP of 727 records (Finding 7). *Allele frequency:* causal alleles are absent/ultra-rare in gnomAD, consistent with strong constraint. *Origin:* germline, typically de novo. *Functional consequence:* loss of function → haploinsufficiency. *Distinct allelic mechanism:* exon-9-skipping *CUL3*-Δ9 → dominant-negative → hypertension disorder ([PMID: 29361671](#), [PMID: 32619053](#), [PMID: 35563538](#)). *Modifier genes / epigenetics:* not specifically defined for NEDAUS; substrate-adaptor genes (*KCTD13*, *KLHL* family) are mechanistic partners. *Chromosomal abnormalities:** 16p11.2 CNVs (containing *KCTD13*) converge on the same *CUL3*–RhoA pathway ([PMID: 37465586](#)).

5. Environmental Information

No environmental, lifestyle, or infectious agents are established as causes or triggers. This is a monogenic disorder.

6. Mechanism / Pathophysiology

Detailed in the Mechanistic Model section and Findings 2, 3, 6, 9. **Molecular pathways:** ubiquitin–proteasome system (*CUL3*), mTOR–eIF4E cap-dependent translation (via 4E-BP1), Rho-GTPase/actin signaling (via RhoA/*KCTD13*), and KEAP1–NRF2 oxidative-stress response. **Cellular processes:** cortical neurogenesis, neuronal migration, dendritic maturation, network-activity formation. **Protein dysfunction:** loss of scaffold function → reduced substrate ubiquitination → substrate accumulation. **Transcriptomic/multi-omic** dysregulation of neurogenesis and cytoskeletal programs is documented in the mouse model ([PMID: 33727673](#)).

7. Anatomical Structures Affected

Primary organ: brain (UBERON:0000955), specifically cerebral cortex (UBERON:0000956) with reduced volume and delayed myelination (HP:0002188). **Body system:** central nervous system (nervous). **Secondary involvement:** cardiovascular (septal defects), craniofacial/palate, and occasionally limb. **Cell types (CL):** cortical projection neurons (CL:0000598), neural

progenitors (CL:0000047), pyramidal neurons. **Subcellular (GO CC):** cytosol (GO:0005829), Cul3-RING ligase complex (GO:0031463). **Lateralization:** bilateral/generalized CNS involvement.

8. Temporal Development

Onset: congenital/infantile (HP:0003593). **Course:** static (non-progressive) neurodevelopmental encephalopathy; epilepsy may be episodic and infantile spasms have a characteristic early-infancy window. **Critical period:** early cortical development (fetal/early postnatal neurogenesis and migration) is the mechanistically vulnerable window, consistent with the mouse MRI finding of decreased cortical volume from early postnatal development. **Duration:** chronic, lifelong.

9. Inheritance and Population

Inheritance: autosomal dominant (HP:0000006), predominantly *de novo*. **Penetrance:** essentially complete for the neurodevelopmental phenotype; **expressivity highly variable**. **Epidemiology:** ultra-rare; precise prevalence/incidence not established (fewer than ~50 published patients). **Sex ratio:** not firmly established; model organisms suggest possible male-biased severity. **Founder effects/consanguinity:** not applicable (dominant, *de novo*). Recurrence risk to siblings is low (germline mosaicism possible but not quantified); risk to offspring of an affected individual is 50%.

10. Diagnostics

Genetic testing is definitive. Recommended approach: trio **whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)**, or a broad intellectual-disability/autism/epilepsy **gene panel** that includes *CUL3*; **chromosomal microarray (CMA)** detects gene/exon-level and 16p11.2 CNVs. Variant interpretation by ACMG/AMP (PVS1 for null variants). **Supportive tests:** EEG (may show hypsarrhythmia/burst suppression), brain MRI (reduced cortical volume, delayed myelination). No specific biochemical biomarker exists, though patient cells show reduced ubiquitin conjugates and elevated 4E-BP1 (research assays). **Differential diagnosis:** other monogenic syndromic NDD/DEE genes (e.g., STXBP1 [PMID: 42609058](#), SCN2A, MED13L, SHANK3, KCNB1, PTEN, CDKL5 — several co-identified in ASD/NDD cohorts, [PMID: 25969726](#), [PMID: 31696658](#)) and 16p11.2 deletion/duplication syndrome.

11. Outcome / Prognosis

NEDAUS is a chronic, lifelong disorder. It is not typically life-limiting on its own, but severe epilepsy (infantile spasms/DEE), feeding difficulty/failure to thrive, and structural anomalies contribute morbidity. **Prognostic factors:** severity of intellectual disability, presence and control of epilepsy, and structural comorbidities. Long-term outcomes reflect the degree of developmental impairment; recovery is limited (static encephalopathy) but developmental gains occur with intervention.

12. Treatment

No disease-specific therapy exists — management is symptomatic and supportive: - **Antiseizure medications** for epilepsy/infantile spasms (standard infantile-spasms therapy: ACTH, vigabatrin, corticosteroids as clinically indicated). - **Developmental/behavioral therapies:** early intervention, special education, applied behavioral therapy for autism, speech and occupational therapy. - **Feeding support** for feeding difficulty/failure to thrive; surgical repair of palatal/cardiac anomalies as needed. - **Emerging mechanistic targets (preclinical):** **RhoA inhibition** rescued dendritic and network phenotypes in the mouse model ([PMID: 33727673](#)); the **mTOR–eIF4E/4E-BP1 translational axis** is a rational second target ([PMID: 39301775](#)). Neither is an approved therapy for NEDAUS. No pharmacogenomic or gene/RNA-based therapy is established.

13. Prevention

No primary prevention exists for a *de novo* dominant disorder. **Genetic counseling** is central: an affected individual has a 50% transmission risk; sibling recurrence risk is low but germline mosaicism cannot be excluded. **Prenatal/preimplantation genetic testing** is available for families with a known pathogenic variant. Secondary/tertiary prevention focuses on early epilepsy control and developmental intervention to limit complications.

14. Other Species / Natural Disease

Cul3 orthologs are conserved: mouse *Cul3* (NCBI Gene 26554) and rat *Cul3* (NCBI Gene 292630). **Model species:** *Mus musculus* (NCBI:txid10090), *Rattus norvegicus* (NCBI:txid10116). No naturally occurring companion-animal disease is catalogued in OMIA for *CUL3*, but the pathway is evolutionarily conserved and disease mechanisms recapitulate across humans, mice, and rats ([PMID: 37465586](#)). No zoonotic relevance (non-infectious genetic disease).

15. Model Organisms

- **Mouse (*Cul3* germline haploinsufficient, CRISPR):** recapitulates social/cognitive deficits, hyperactivity, reduced cortical volume, and dendritic/network deficits; RhoA inhibition rescues key phenotypes — strong construct and face validity ([PMID: 33727673](#)). Databases: MGI, IMPC.
 - **Rat (16p11.2 deletion/duplication, outbred SD & LE):** altered MAPK1/CUL3 pathways, social-behavior deficits, sexual dimorphism ([PMID: 37465586](#)). Database: RGD.
 - **In vitro / patient-derived cells:** patient T-cells demonstrate reduced ubiquitin conjugates and 4E-BP1 accumulation, providing a cellular readout of the mechanism ([PMID: 39301775](#)).
 - **In utero electroporation (mouse):** KCTD13/TNFAIP1 manipulation alters cortical neuron positioning and dendritic maturation ([PMID: 26969432](#)).
 - **Limitations:** models capture cytoskeletal/neurogenesis phenotypes but incompletely model the full human cognitive/autistic/epileptic spectrum; iPSC-derived neuron/organoid models of NEDAUS are not yet established in the reviewed literature.
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Evidence Base

PMID	Title (abbrev.)	Role in this report
39301775	<i>Loss-of-function variants in CUL3 cause a syndromic NDD</i>	Landmark human cohort (n=37, 35 LoF); defines phenotype + 4E-BP1 mechanism (F002, F009)
33727673	<i>Autism-linked Cullin3 germline haploinsufficiency...RhoA signaling</i>	Key mouse model ; cortical volume ↓, RhoA rescue (F003)
37465586	<i>...16p11.2 rat models...MAPK2 and KCTD13/CUL3</i>	Cross-species conservation; CUL3-pathway alteration (F008)
29361671	<i>Mutant Cul3-mediated familial hyperkalemic hypertension</i>	Defines distinct CUL3-Δ9 dominant-negative allelic disorder (F004)
26969432	<i>Bacurd1/Kctd13 and Bacurd2/Tnfrsf1...Rnd proteins</i>	Adaptor→RhoA/cortical positioning mechanism (F006)
32619053 , 35563538 , 27378813	PHA II / CUL3-Δ9 papers	Support the distinct hypertension mechanism
42446772 , 37328017	KEAP1–NRF2 / Cul3 antioxidant papers	Supporting CUL3 substrate network (F006)
31696658 , 25969726	ASD/NDD WES cohorts	Independent identification of <i>de novo</i> CUL3 LoF in ASD; differential-diagnosis context
31279627	<i>Structural basis...KLHL20 E3 ligase</i>	Structural context for BTB-Kelch CUL3 substrate recruitment

Evidence-source types: human clinical cohorts (39301775, 31696658, 25969726); model organism (33727673, 37465586, 26969432, 27378813); in vitro/patient-derived cells (39301775, 35563538, 37328017); population genetics/computational (gnomAD constraint, ClinVar).

Limitations and Knowledge Gaps

1. **Small case numbers.** The largest cohort is n=37; precise prevalence, incidence, sex ratio, and natural-history milestones are not established. Phenotype frequencies for several HPO terms rest on small denominators (e.g., 2/3, 3/3).
 2. **Genotype–phenotype correlation is underpowered.** The basis for highly variable expressivity (why some individuals have autism, others seizures, others neither) is unexplained; no confirmed human modifier genes or epigenetic modifiers.
 3. **Two missense variants** in the cohort are not fully mechanistically resolved versus LoF — the contribution of non-null alleles to NEDAUS remains uncertain.
 4. **No human neuronal disease models** (iPSC-derived neurons/organoids) were identified in the reviewed literature; mechanistic inferences rest on mouse/rat and patient blood cells.
 5. **No therapeutic trials.** RhoA and mTOR–eIF4E rescue data are preclinical only; translational safety/efficacy in humans is unknown.
 6. **Biomarkers** (4E-BP1 accumulation, ubiquitin-conjugate levels) are research assays, not validated clinical diagnostics.
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Proposed Follow-up Experiments / Actions

1. **Establish an international NEDAUS registry** to define prevalence, natural history, epilepsy trajectories, sex ratio, and genotype–phenotype correlations at scale.
2. **Generate patient iPSC-derived cortical neurons/organoids** to test whether 4E-BP1 accumulation and RhoA-dependent cytoskeletal/network phenotypes reproduce in human neurons, and to screen rescue compounds.
3. **Preclinical target validation:** test RhoA inhibitors (e.g., ROCK-pathway or direct RhoA modulators) and mTOR–eIF4E/4E-BP1-axis modulators in the *Cul3*^{+/−} mouse for behavioral/cognitive rescue and therapeutic window.
4. **Deep phenotyping of the CRL3 substrate/adaptor network** (KCTD13, KLHL family, NRF2/KEAP1) in patient cells to identify secondary biomarkers and additional therapeutic nodes.
5. **Systematic reclassification of ClinVar VUS** (of the 727 records, ~500 non-P/LP) using functional 4E-BP1/ubiquitin-conjugate assays to improve diagnostic yield.

6. **Distinguish allelic disorders diagnostically:** ensure clinical pipelines flag exon-9-skipping CUL3-Δ9 variants (hypertension) separately from whole-gene LoF (NEDAUS), given opposite mechanisms and management.
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Report compiled from 9 confirmed findings and 27 reviewed papers over 5 iterations. Evidence is strongest for the core identity (F001), haploinsufficiency mechanism with 4E-BP1 accumulation (F002), and RhoA-dependent cortical/behavioral phenotypes with pharmacologic rescue in mouse (F003).

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